Data center security

When you tour a HPC AI data center with special forces operators, they ask some weird questions. Who manufactures the fire alarms? [redacted] Where were the critical mineral inputs to the AI accelerators processed? What construction company was contracted to build the facility? The tactics, techniques, and procedures (TTPs) actually available to and employed by our nation-state adversaries are more advanced than hyperscalers, private labs, chip design firms, and data center infrastructure providers can know. That will have to change if a U.S. government superintelligence project is going to succeed.

> *With a [redacted], what you do is you get [redacted], and you knock out the whole [data center component] with [redacted] You just rip it across the [component] and fucking take it out.*[11]
>
> – Former special forces operator evaluating the security of an AI data center housing hyperscaler infrastructure

To train and deploy frontier AI models, you need AI hardware to run computations and move data around, a reliable source of high base load power to keep your hardware running, and cooling systems to stop your hardware from overheating. If you want to truly secure a U.S.-backed superintelligence project, then you need to secure the supply chains for these assets – starting with the data centers themselves.

Data centers are where the critical supply chains for AI chips, power, and cooling come together. They're the facilities in which a U.S.-backed superintelligence project will actually train and run its models. Because of their size and complexity, data centers have lots of human, physical, and cyber security access points and vulnerabilities. Nation states can easily exploit those vulnerabilities to extract sensitive IP or to damage key facilities, in ways that aren't addressed by any known current or proposed future security measures.

In November 2024, we joined a team of former Tier 1 special forces operators and national security intelligence professionals and were given access to a [redacted] data center containing

[11] Obviously this is a very redacted quote. But the attack described could allow someone to knock out a >$2B facility on a sub-$30k budget so we were specifically asked not to disclose it in too much detail. What's worth noting here is how quickly and casually individuals with experience executing nation state-level operations on similar facilities can identify these extreme asymmetric vulnerabilities.

hyperscaler AI clusters, in order to assess its security posture. We later also spoke with individuals directly responsible for the security of hyperscaler-owned and operated AI data centers.

Our goal was to understand the current state of data center security, to determine how it needs to be improved to resist nation-state attacks, and whether we can expect those improvements to happen in time for ASI-grade training runs if they were executed in the 2026-28 era. If we take seriously the prospect that superintelligence will be a WMD and or WMI (weapon of mass influence), then these data centers will become priority targets for the intelligence and special operations of nation-state adversaries and their covert surrogate networks. So in order to understand how to secure a superintelligence project, we need to talk to intelligence and special forces operators directly.

Physical security

Today, security practices vary widely between AI data centers operated by hyperscalers[12] and colocation providers, but some trends are clear. A given data center tends to have a highly uneven security posture across different threat vectors. For example, it's often hard for employees or visitors to take photographs of data halls, but it would be easy to [redacted]. This inconsistent security profile is a problem. Even the most secure windows in the world don't matter if you leave the front door open.

By the assessment of one special forces operator with direct firsthand experience running [redacted] operations in front-line environments, a strike capable of destroying a key component of a hyperscaler data center infrastructure "could be done with [redacted] for under 20k [dollars]. Pending you had all the necessary component parts to [redacted]." When we asked the data center's security lead how long it would take for them to get the facility back online after a strike like this, his reply was "A year?

[12] There's no consistent frameworks for data center security today, so security measures reflect individual companies' tolerance for risk. This is because the security requirements that data center builders and operators follow typically come from contractual obligations to the data center's users and not from any particular set of best practices. As it becomes increasingly obvious that frontier AI models are key national security assets, this will have to change. Executive and/or legislative actions will have to create more consistent and stringent requirements on hyperscalers building and running inference on frontier systems.

Six months at least. [redacted]"[13]

That's a $2B facility out of commission for over 6 months, on a budget of $20-30k. All from a threat that's been completely overlooked by every widely referenced assessment we're aware of. Fortunately, it's also a vulnerability that's easily fixed: according to one special forces operator, the lowest-cost [redacted] variants of this attack could be blocked by simple and cheap add-ons.

But this also wasn't the only critical [redacted] vulnerability the team identified. Many critical components of data centers are on multi-year back-order and are highly exposed even in the most secure builds. As one operator explained, "With a [redacted], what you do is you get [redacted], and you knock out the whole [data center component] with [redacted]. You just rip it across the [component] and fucking take it out." We've been asked not to share the specific components or techniques involved in these attacks precisely because of how significant these vulnerabilities are, and how cheaply they can be exploited.

These low-cost, highly leveraged and disruptive attacks could easily prove decisive in the context of a U.S.-government backed project designed and understood to be aimed at developing weaponizable superintelligence, among other capabilities. In particular, if we find ourselves in a race to superintelligence with the CCP, a cheap attack on an unprotected data center could set our efforts back by months or even years depending on supply chain delays.[14] And although special authorities could considerably reduce wait times for key components, significant downtime for core data center infrastructure could still be decisive in an adversarial race.

Recommendation: At the design stage, **new data center projects that are aiming to support training and inference for frontier models by the end of 2025 should enlist support from individuals with direct experience performing nation-state denial operations,** including but not limited to operations involving [redacted]. These should specifically include red teaming, pentesting, and tabletop exercises supported by current and former special operators to address vulnerabilities introduced by new platforms as they come online during design and construction. This activity should be coordinated with the U.S. government, and involve the

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[13] [redacted]

[14] Many of these critical components are themselves sourced from China, which brings its own major vulnerabilities that we'll get into in later sections.

development of a variant of the [redacted] We can provide further information upon request.

The IC and national security community will need to work with data center builders, hyperscalers, and frontier labs to define security measures that can withstand attacks by resourced nation state adversaries. That should include funding or incentivizing fast prototyping build-outs and retrofits of data centers and supporting infrastructure that can withstand those kinds of attacks. Some or all of these projects will probably fail to meet the full level of security they need. Funding for these prototype builds will have to be risk-tolerant the same way venture capital is: we can't let fear of failure hold back critical experiments in data center security.

We can start with small-scale prototypes. For example, an individual GPU or a pod of 8 GPUs with simulated network fabric is something that can be set up very quickly. It can also be red teamed and pentested fast before it's scaled up to a bigger build that fixes the security problems that get discovered in smaller-scale iterations.

Ultimately, perfect security is impossible, and we shouldn't expect to be able to build an impenetrable fortress. But **we should raise the bar for security to the point where our adversaries' attempts to take down key frontier AI infrastructure are attributable**, so that these attacks lead to a credible threat of retaliation.

Hardware supply chain security

According to the AI data center builders and operators we spoke to, the supply chains they'd need to draw on to provision secure physical data center infrastructure are badly strained. Those constraints will impact training runs and deployments of leading AI models in the 2026 era and beyond. Most at risk are components whose supply chains depend heavily on Chinese manufacturers, and which in some cases are produced exclusively by Chinese companies. To take just one example, the overwhelming majority of transformer substations contain components that were made in China and can be used as back-doors for sabotage

operations.[15] Indeed back-door electronics are known to have been installed in Chinese-made transformers. If a superintelligence project were kick-started under nominal conditions, unsecured supply chains for AI hardware, as well as electrical and cooling infrastructure could embed physical CCP trojan horses deep into the data centers that house some of the most national security-critical technology America will ever build.[16]

What's more, China announced a $275B investment in AI infrastructure[17] within days of the announcement of OpenAI's Project Stargate. In this context, we'd be naive to expect that China is going to prioritize shipping critical data center components to U.S. AI labs over its own directly competing projects. Therefore we should expect that the lead times on China-sourced generators, transformers, and other critical data center components will start to lengthen mysteriously beyond what they already are today. This will be a sign that China is quietly diverting components to its own facilities, since after all, *they control the industrial base that is making most of them*.[18]

As one example of foreign supply chain dependence, a critical and underappreciated security problem for AI data centers is the supply chain for Baseboard Management Controllers (BMCs). BMCs are microcontrollers that sit on server motherboards and handle key housekeeping tasks, like monitoring temperature, voltage and power flows, detecting memory errors, and monitoring network connectivity. BMCs operate even when the main server they monitor is powered off, and they have a privileged position in the system architecture: among other things, they can directly interface with baseboard hardware, and they can read the firmware that links the baseboard hardware to the operating system. This means that attackers who gain control of the BMC can [redacted]

[15] Other examples include transformers, generators, uninterrupted power supplies, copper cables, etc.

[16] This isn't just a problem for the initial build of new data centers and supporting infrastructure. New components are in constant demand due to the need for continuous maintenance, upgrades, and replacement of failed components.

[17] The actual amount of the investment is 1 trillion yuan. This has been frequently reported as a $137B investment, but in reality it's closer to $275B when measured in purchasing power parity (PPP) terms — since that's a better reflection of what you can buy in-country.

[18] If they're clever about it, they will of course never announce these diversions as a policy — we'll just see lengthening delays and chalk them up to increased demand, competitive bidding for rush orders, mundane shipping problems, and so on.

[redacted][19]

BMCs are a "soft underbelly" of AI server security – and a dream target for adversaries looking to Stuxnet some of our most critical national security assets. Which is why it's especially bad that Taiwan-based ASPEED makes 70% of the world's BMC chips.[20] ASPEED BMCs are all over the place, appearing in every Nvidia DGX server – and, while they're modular in more recent Blackwell systems, they're being widely deployed in hyperscaler AI data centers. We should expect ASPEED to already be a target for the CCP, and for their ASPEED infiltration, compromise, and sabotage operations to be increasingly prioritized as the stakes of superintelligence get more widely recognized. We need made-in-America alternatives.

Beyond BMCs, many of the less "sexy", lower-tech but essential infrastructure components that make data centers hum are key vectors for adversary operations. Programmable logic controllers (PLCs) that power critical power and cooling systems, for example, have more diversified supply chains, but are also under-appreciated attack channels.

Reshoring critical supply chains to America or our allies won't be a matter of months, but of years. We won't have time to do it after we've come to a political consensus around securing a superintelligence project. But whether or not we ultimately choose to kick off a superintelligence project, if we want our infrastructure to be robust to even the most basic of these supply chain attacks, we need to start now.

There have been proposals to screen physical hardware that enters data centers as part of supply chain security. While this helps, it's not adequate against a resourced attacker that controls major parts of the supply chain, as China does. We can't go into details, but we were informed during our investigation that there exist techniques for supply chain sabotage, including for destruction and surveillance, that are [redacted]. As a former intelligence official informed us, the only guaranteed defense is full control of the supply chain from mineral extraction to final assembly

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[19] Based on private communication with Seth Demsey.

[20] Notably, ASPEED BMCs are starting to integrate open-source RISC-V cores which have not been properly vetted for security. Open-source software in general has security upsides, since in principle it's auditable by external, security-minded developers. Unfortunately the ASPEED RISC-V BMC has received very limited security scrutiny to date, particularly relative to proprietary alternatives from Arm and Intel. Chinese entities have invested heavily in supporting the rapidly growing RISC-V ecosystem.

and installation of every component. Of course, perfect security is impossible, and security standards have to be balanced against development velocity in the context of a technological race. But the intelligence official who shared this assessment with us was basing it on an overall evaluation of the risk profile of other forms of WMD development: if we're going to treat superintelligence like the weapon that it will be, then that's just the security bar we have to meet.

Recommendation: We need as much production capacity associated with critical data center components – like transformers, power supplies, cables, networking equipment, and cooling components – onshored as possible. We know from conversations with national security professionals that there are ways to doctor components in the supply chain for surveillance or sabotage that [redacted]. If adversaries are manufacturing most of our critical components, that's a problem we need to solve immediately. One possible workaround is [redacted] We need to assess how feasible this would be for each critical component in the AI supply chain so that we can pull the trigger on this option if we determine that onshoring manufacturing of these components will take too long.

We also specifically need a securely sourced alternative to ASPEED BMCs — ideally, a simple, purpose-built and easily auditable drop-in replacement for existing chips. DARPA projects focused on developing secure BMC hardware could be a helpful step towards a solution, but government purchase guarantees could drive demand in tandem. To mitigate risk in the shorter term, [redacted]; insist on having suppliers share information about their sourcing of BMC chips; and introduce requirements for supplier diversity, including at the firmware level. [redacted]

These would be big — and expensive — moves, but they reflect the scale and impact of the vulnerability they're meant to address. BMCs offer our adversaries a way to conduct crippling attacks on our most important compute infrastructure. We can share extensive further recommendations if desired.

Side-channel security

Adversaries will almost certainly aim to knock out the American data centers powering training runs for a superintelligence project. But to secure the project, we'll need to do more than prevent adversaries from destroying AI infrastructure – we'll also need to prevent them from **stealing critical IP**.

A widely-cited report by AI security researchers published in 2024 provides a reasonable starting point for understanding what it would take to secure AI model weights from theft by nation-state attackers. It defines AI systems requiring Security Level 5 (SL5) as systems that would become targets for "top-priority operations by the top cyber-capable institutions" like the People's Liberation Army's Strategic Support Force, China's Ministry of State Security, and analogous institutions in other adversary countries. Superintelligence could offer those who control it a decisive strategic advantage, and would almost certainly need to be secured at the SL5 level. The recommendations at this level of security are extensive. They include:

- Strict limitation of external connections to a completely isolated network
- Routine rigorous device inspections
- Disabling of most communication at the hardware level
- Extreme isolation of AI model weight storage (completely isolated network)
- Advanced preventive measures for side-channel attacks (e.g., noise injection, time delays, and other tools)
- Formal hardware verification of key components

To quote one former Mossad cyber operative, achieving the proposed SL5 standard would be "really hard. Even people who've read the report and understand cybersecurity underestimate how hard this would be." In practice, SL5 would require U.S.-made components, or highly trusted components that the Department of Defense or the Intelligence Community (IC) would be comfortable using for extremely classified operations.

But as we mentioned earlier, many of the components in current AI clusters are sourced directly from China. As one cyberoperative put it, current clusters are, "not even close to being close. No one is even thinking about this. This is not a priority for them. [...] AI companies have funds, but not for this; people who are smart, but not on this topic. It's just not in their skillset. [...] Government could do this, sure, but this is a big project; this is a mega-project. I'm not [aware of] ultra-secure projects that would be this size, and require this much specialized equipment at the drop of a hat."

And the report's recommendations aren't even comprehensive: they leave out key security challenges associated with geographically distributed training, for example. They also only address the risk of model weight *theft*, and not attacks meant to disable or destroy computing infrastructure, which, as we've seen, would be extremely cheap and damaging under current conditions.

These aren't long-term problems either. An AI security expert who collaborates frequently with frontier AI labs estimated that the next generation of AI models could already qualify for the SL4 threshold – defined in the report as security measures required to counter "most standard operations by leading cyber-capable institutions." By his assessment, even SL4 security measures would take at least 1.5 to 2 years to implement under current conditions, requiring a "massive investment" from key players. This individual assessed that the shift from SL4 to SL5 security would also be significant, almost certainly requiring entirely new data center facilities built from the ground up – a perspective echoed by several other physical and personnel security experts we spoke with.[21]

Our interviews with energy and compute infrastructure specialists, intelligence experts, and special operations personnel have surfaced other critical challenges. For one, data center construction can take well over a year to complete, and data centers probably can't be retrofitted to meet these criteria after they've been built. Similarly, the AI accelerators that will fill frontier AI data centers in 2026 and beyond are having their designs finalized. They will have to be redesigned to address accelerator-level vulnerabilities that will be relevant in the geostrategic and security environment in coming years, but which aren't on leading players' radar today. Absent unprecedented government incentives driving a major security push in the very near term, secure AI accelerators will not be available in time for a superintelligence project.

One key attack vector is **side-channel attacks**, which include things like Telecommunications Electronics Materials Protected from Emanating Spurious Transmissions (TEMPEST) threats.

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[21] By contrast to SL4 and SL5, which are security thresholds intended to be capable of thwarting nation-state operations, SL3 is in principle designed to defend against operations by non-state actors such as cyber crime syndicates, terrorist organizations, or insider threats. The SL3, SL4, and SL5 thresholds are widely referenced in the AI security community, but intelligence officials we spoke to made it clear that in practice, non-state actors are often trained or even backed by states, and can use advanced TTPs that most analysts would associate with "nation-state" capabilities as a result. So in practice, and contrary to conventional wisdom in the AI security ecosystem, dealing with threats from non-state actors will require many measures drawn from the SL4 and SL5 standards.

These are information extraction attacks that use leaked electromagnetic, sound, or vibrational signals to reconstruct sensitive and intelligence-bearing information.

It's been argued that TEMPEST attacks would not be able to exfiltrate model weights themselves due to their limited bandwidth, but that they could be used to extract decryption keys or other metadata that makes model weights easier to access or reconstruct. However, our investigation suggests that TEMPEST technologies may be far more advanced than is widely recognized. [redacted] We assess that it is plausible that some adversary nation-states have access to TEMPEST and other side-channel attacks that could allow for direct weight exfiltration under certain conditions.[22]

Although many are classified, some standard defenses against TEMPEST attacks include ensuring a minimum distance between HVAC conduits and sensitive equipment, using shielding and Faraday cages, implementing physical security control zones around sensitive areas, using fiber optic cables instead of copper cables where possible,[23] and air-gapping critical components. Many of these strategies have to be accounted for at the design stage of a new data center build; it won't be possible to retrofit existing facilities to meet appropriate TEMPEST standards. Failing to do so would bake critical vulnerabilities into a government-backed superintelligence project.

Recommendation: At the design stage, and throughout the build, **new data center projects aiming to support training and inference for frontier models by the end of 2025 should work directly with intelligence and national security agencies to identify and mitigate vulnerabilities from TEMPEST and other side-channel attacks.** This work should also involve current or former frontier AI researchers, hardware design specialists from leading companies like NVIDIA/AMD, and data center infrastructure design experts.

Only the U.S. national security and intelligence communities, with support from private sector experts retained at leading chip design, data center, and AI model development companies,

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[22] As one indication that we can share: we were shown a live demonstration of an attack that allows hackers to reconstruct the architecture of a small AI model using nothing but the power consumption profile of the hardware that runs it.

[23] However, our discussions with former cyber operators suggest that even dark fiber connections between geographically separated data centers are highly vulnerable to established nation-state attacks, the details of which are not public.

are in a position to estimate the actual extent of the TEMPEST and cyber threats that a superintelligence project would face. And this holds for vulnerabilities beyond TEMPEST attacks too: there almost certainly exist highly classified methods, technologies, and tools available to adversary nation states whose capabilities will have to be accounted for from the earliest design stages of a superintelligence project.

There is no other option: the IC and national security community need to be involved in project design and in the design of its supporting infrastructure – *including of the AI accelerators themselves* – from the very beginning. As one former military security specialist put it, "I wouldn't consider [AI accelerators nation-state proof], unless the NSA or CIA is involved in designing them."

Software supply chain security

The software stack that frontier AI labs use, and that runs on AI data center infrastructure, is insecure. According to a cybersecurity specialist with experience executing nation state cyber operations, "There are at least several incidents you could point to with libraries that are used by leading labs that have been compromised in the past, whether by the intelligence communities of other countries or accidentally." In addition to these known vulnerabilities, nation-state adversaries have almost certainly identified zero-day vulnerabilities.

In this respect, the CCP is one among many potential threat actors. Although China's physical access to sensitive supply chains makes it the leading threat actor along that vector, Russia has shown itself to be highly capable in the cyber domain. And while Russia lacks the AI hardware and talent it would need to train frontier AI models, they may have enough infrastructure to *run* a frontier model if they steal one in a cyber operation. In a world of compromised frontier lab security, **cyber powers *are* AI powers.**

> *No one is doing this, no one is looking into this, no one is trying to create an end-to-end secure software stack that is secure – not just in quotation marks – but genuinely secure.*
>
> – Former IDF cyber security operator on frontier AI lab cyber security

These challenges create a dilemma: nation states aren't pricing in the military implications of superintelligence, which makes it difficult to justify investing in basic security measures today.

But lead times for energy infrastructure components are measured in years, chip design cycles are even longer, and security features can't be retrofitted into hardware, so delay is not an option. We can either place a bold bet on securing our critical AI infrastructure today, or find ourselves racing the CCP to superintelligence with an easily-penetrable hardware and software stack.

Recommendation: There are lists of software libraries that are approved for use in military systems at various levels of classification.[24] **We should open up and expand this process for deep learning-supporting software libraries used by frontier labs, and resource it so it can move as fast as it needs to**. This probably also requires collaboration between the research engineers and the accreditation orgs, as well as the involvement of the intelligence community. The speed of this review process should be explicitly balanced against the need for rapid development, with the understanding that a single critical breach could cause the project to fail.

Energy security

Energy production is an often-cited bottleneck to frontier AI development, and increasing high base load power availability on the U.S. electric grid should be a top national security priority. We won't rehash the details here since there are many public reports on this already, other than to say that our interviews with energy specialists suggest that new natural gas plants capable of supporting hundreds of megawatts currently take 5 to 7 years to set up, but could be built in just over two years – quickly enough to support training runs of the 2027–2028 era – if permitting and supply chain bottlenecks were removed.[25]

The speed at which new natural gas plants can be brought online (assuming we remove regulatory frictions) makes them a particularly promising way to support a U.S.-backed superintelligence project in the near term. Other options like nuclear and geothermal matter more on longer timescales.

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[24] For example, Iron Bank sets and verifies standards for hardened containers used in software development for sensitive DOD applications.

[25] This is worth emphasizing: it means that regulatory hurdles are currently the primary reason that new natural gas capacity could not be brought online in time to support a 2027/28 superintelligence project.

Gas turbines are already sold out across the market, and other components are on long-term back-order. This won't be fixed in time to make a difference for the training runs of the 2027 era. U.S. government policy can move the needle here though, as we'll explain below.

From speaking with cabinet officials in U.S. states, a less recognized problem is the time cost of litigation over environmental concerns. Litigation creates a huge bottleneck to new power and infrastructure build-outs. Strategic economic interference by China is an important factor here. We've heard from former counterintelligence and special operations personnel that Chinese entities are explicitly funding and leveraging domestic environmental groups to tie up data center and other critical national security builds through litigation. The environmental groups involved in this work are not generally aware of the ultimate source of their funding — the funding is often indirect and funneled through proxies and surrogates — but effectively operate in alignment with adversary interests.

Recommendation: We know it's possible to build *massive* amounts of power generation quickly — up to 3-5 GW or more[26] — as long as we're willing to ignore carbon emissions concerns in the short term and just set up huge gas turbines.[27] In terms of energy, we're more bottlenecked by our own regulations than by economics or physics. In practice, we could pick a site for a project, exempt it from regulations, and probably build as much power as we need there quickly. To help resolve supply chain problems, [redacted]. Over time we could even phase out gas turbines and replace them with lower carbon sources, but there's no need for that to delay other parts of the project.

Deregulation is essential, and plenty of people are already discussing it. Environmental regulations are a particularly big problem here, and the President could invoke the DPA Title III

[26] Enough to power about 2-3 million American homes, or the state of Utah. It's around 1-2% of the power generation capacity of the entire United States. At this scale you can power about 2 million H100 GPUs (depending on various assumptions), equivalent to at least 10x more compute capacity than we believe exists at any one site as of January 2025.

[27] The Department of Energy didn't actually think this was possible when we spoke to them in December 2024: they told us there was no way to pull together more than 1 GW of capacity in any one place over the next couple of years. But in fact it's exactly how Elon Musk was able to get xAI's Memphis data center up and running in record time. The 3-5 GW number was also confirmed to us by a couple of people who are actually working on deploying those gas turbines. Our best guess is that the DOE team we spoke to must have been assuming only low-carbon energy sources.

to provide NEPA exemptions[28] to advanced AI clusters and related dedicated energy infrastructure that meet various security requirements. We also need categorical exclusions to NEPA for no-impact preliminary activities on these data center projects, like design, drilling boreholes, and collecting soil samples. This would unlock early DOE financing through loans.

We should direct the creation of limits on litigation options for designated critical infrastructure projects, and in particular, expedited court decisions. **Every delay mechanism that could be used to tie up new data center and related energy projects should be bounded.**

Export controls

Just as we need to ensure robust supply chains for our own efforts, we need to deny the same to our adversaries. While U.S. GPU export controls have strengthened over the past few years, Chinese firms are known to have subverted controls to acquire controlled chips manufactured and packaged by TSMC by simply spinning up subsidiaries and shell companies that don't appear on entity lists. To take just one example reported by SemiAnalysis, leading Chinese fab SMIC operates two factories linked together by a wafer bridge, which effectively makes them a single facility, yet they were treated as separate entities under then-current U.S. export control policy — one was included on the U.S. entity list, and the other wasn't.

Recommendation: We'll have to move from entity blacklists to selective entity whitelists, significantly limiting the access that photolithography and optics companies, chip fabs and design firms currently have to the Chinese and other markets. We'll also need to pursue less targeted export bans: current highly targeted approaches aimed at restricting chips based on narrow technical specifications still allow China to import significant volumes of strategically critical AI processing hardware. Recent updates to export control rules have tried to address this problem, but have left major loopholes.[29] China has repeatedly demonstrated their ability to integrate and use heterogeneous and legacy hardware into significant training clusters, and

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[28] However we should note that according to a White House official with experience writing executive orders that include NEPA exceptions, they can be legally challenging.

[29] For example: recent AI diffusion regulations limit certain non-allied countries to importing no more than 50,000 advanced GPUs. However, chip orders of under 1,700 GPUs do not count towards this cap. This creates an obvious loophole that is precisely aligned with the demonstrated strategies and capabilities of the CCP, which has historically imported large quantities of otherwise controlled AI chips via a network of subsidiary companies, often spun up specifically for that purpose.

it is unlikely that even the best-informed technical assessments available to the BIS (the U.S. office that manages export control compliance) will be able to anticipate the workarounds they'll develop to any set of narrowly scoped controls.

AI hardware security

A national superintelligence project would need to secure a lot of different supply chains, or [redacted]. But in addition to taking a "wide" view, and solving for security across many devices and pieces of infrastructure, it would also have to take a "deep" view, and consider security exposure from geographically distributed supply chains for logic and memory. If the project depends on chips that are being made in Taiwan and South Korea, then securing and defending TSMC and SK Hynix becomes America's de facto responsibility. All it takes is for the CCP to hack into the right network and modify the firmware that goes on fabricated chips, and the project can be compromised.[30]

In terms of AI GPUs — the specialized processors that actually train and run AI models — an obvious big concern is that most AI GPUs are made by TSMC in Taiwan, and that China claims Taiwan as its own. There's a long history of illicit technology transfer from TSMC to mainland China, and TSMC is likely heavily infiltrated by CCP spies and saboteurs.[31]

There are a lot of hard steps involved in building a GPU, and most of those steps happen at TSMC facilities in Taiwan today. Two of the most important are *fabrication* and *packaging*. Fabrication is where you use lasers and chemicals to etch the tiny circuits into the chip[32] that implement its logic. Packaging is where you combine chips together on one board in such a way that they can talk to each other and with other components like memory and power.

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[30] There may be an alternative, however. Trustless computing schemes – such as flexHEG-based devices and protocols – may allow us to use certain hardware components securely even if the components themselves have been tampered with in certain ways. This is one reason why secure computing should be a top national security priority: currently, all frontier AI logic dies are fabbed by TSMC on Taiwanese soil. We should not base our national security strategy on the assumption that the CCP has failed to achieve what must be one of its top national security priorities: infiltrating and compromising TSMC.

[31] SMIC, China's leading semiconductor fab, was founded by Richard Chang, a former senior executive at TSMC. In a series of 2003-2004 lawsuits, TSMC alleged that Chang, as well as several other early SMIC employees, stole critical TSMC IP and used it to massively accelerate SMIC's development.

[32] Really this is called a die, not a full chip. When we talk about fabrication here we mean the step that goes from wafers to dies.

We could fabricate quite a lot of AI chips in America today if we wanted to. TSMC's Fab 21 in Arizona alone likely has the potential to produce about 11.5 million H100-equivalents per year today[33] (a type of GPU),[34] which is already much more than you could power in a 5 GW cluster.[35]

The problem today is *packaging*. Right now TSMC's entire packaging capacity for AI GPUs is equivalent to about 11 million H100s per year,[36] although they're expanding that quickly. Unfortunately almost all of that packaging capacity is located in Taiwan itself.[37] Intelligence experts have warned us privately that there's a significant amount of low grade sabotage being conducted by adversaries on U.S. soil *today*, but it's even easier for those adversaries to disrupt or block supply chains that originate in Taiwan — and harder for us to harden them — than if they were based in America. In the event of a CCP invasion of Taiwan, packaging capacity constraints – in addition to the collapse of logic die fabrication capacity – could quickly become critical.

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[33] As of early 2024 Fab 21 was expected to run about 20,000 wafer starts per month through the facility, at yields similar to those at TSMC's other facilities which have been reported elsewhere as around 80%. That means they're getting 20,000 x 80% = around 16,000 usable dies out the other end each month, or around 192,000 dies per year. NVIDIA and TSMC likely get around 60 H100 chips out of each wafer, which ballparks to around 11.5 million H100s per year if all of Fab 21's capacity were dedicated to doing just that. Right now Fab 21 is mostly making iPhone chips.

[34] We're discussing H100s here, rather than the newer B100/B200 GPUs that are rolling out now. This is to get a conservative lower bound on what's needed. The B100/B200 series can do more computations and also uses less power per computation than the H100 does.

[35] One H100 GPU consumes about 1340 W in the common HGX configuration, and data center power efficiencies typically range between 75-90% (1.1-1.3 PUE). So 11.5 million H100s consume at most around 11.5 million x 1340 W x 1.3 = about 20 GW.

[36] TSMC's Chip-on-Wafer-on-Substrate (CoWoS) packaging capacity was expected to be around 15,000 wafers per month or 180,000 wafers per year by the end of 2024. If we assume 60 H100 chips per wafer, that means almost 11 million H100s can be CoWoS packaged across all of TSMC's facilities.

[37] Fortunately, that is starting to change. TSMC and Amkor recently announced the development of two packaging facilities on U.S. soil, one of which will be located in Arizona and will have a capacity of 14,500 12-inch wafers per month (the second facility's capacity is not yet publicly known). Construction of the Arizona packaging facility is only expected to be complete in 2028, however, by which time it would represent ~10% of TSMC's target 2028 capacity of 150,000 wafers per month. This timeline is too long to address the risk of a collapse in America's advanced packaging supply in the event of a CCP invasion of Taiwan pre-2028.

Upstream of TSMC, there are also vulnerabilities at the level of the photolithography supply chain, which makes critical components TSMC uses to etch nanoscopic circuits onto AI chips. That includes the ASML/Carl Zeiss complex of companies that manufactures the machines that use intricate ultraviolet light sources and optics to pattern circuits onto the chips, and the specialized optical components that go into those machines.

Although it can take years for a new generation of photolithography machines to start being used for high-volume manufacturing of leading edge chips, existing photolithography machines are so complex that they need a full-time team from ASML just to maintain them in good working order. That means adversaries who wanted to limit our access to advanced AI chips could also try to do it at the level of ASML — although that scenario matters most if we expect to achieve superintelligence later than 2027 or 2028.

Recommendation: We should prioritize localizing CoWoS and CoWoS-like packaging, including CoWoS-L and CoWoS-R, in the United States, possibly by extending purchase guarantees to domestic companies like Intel and Amkor.[38] We also need to make sure that the components needed to manufacture chips, such as photolithography machines, are secured. This could involve closer collaboration between the United States and Dutch, Japanese, German, and other governments on counterintelligence, counterespionage, and countersabotage.

We also need to accelerate work on on-chip security technologies like confidential computing at scale, which are going to be critical to any effort to secure frontier model weights. We should be funding and incentivizing chip fabs and design firms to do that work directly, and engaging national security agencies in pinning down the security features that are most badly needed based on the current threat landscape.

A key priority also needs to be establishing [redacted]

[38] Intel already has at least 5000 wafers per month of CoWoS-like packaging capacity (via their Foveros process) — much of which is already located in North America — and is already using it to package NVIDIA H100 GPUs. This is equivalent to 60,000 wafers per year or over 3.5 million H100s at 60 H100s per wafer.

[redacted]

To be clear: some of these infrastructure pieces might not matter at all if we reach superintelligence within the next two years or so. If we get superintelligence that fast, then we probably have enough AI chips in the United States today to train a superintelligence already. And that means that ensuring a domestic supply of AI chips today matters somewhat less, although denying AI chips and developing retrofittable secure computing add-ons (e.g. flexHEG) still matter a lot.

Finally, if all this turns out to be overblown and we're actually decades away from superintelligence, it's important to realize that all these infrastructure expenses won't have been wasted. Even if AI never gets any better than it is today, there is already massive demand for AI inference that can be served more cheaply by bigger data centers, more chips, and more power. At worst, we end up in a situation similar to the dot-com bust, where the fiber optic infrastructure that got built during the boom ended up driving the growth of the modern Internet.